Measurement Strategy for Horizon-Based Optical Navigation in Deep Space Missions

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***Abstract*—Optical Navigation (OpNav) is a powerful tool for spacecraft state determination, encompassing orbit, attitude, and position estimation. In GPS-denied scenarios, it becomes one of the few viable methods for state estimation, making it essential for deep space missions. One sub-method of OpNav is horizon- or limb-based sensing of celestial bodies, in which images of a body's edge or limb provide relative positional knowledge of the spacecraft. When paired with advanced dynamical filtering, such as an Extended Kalman Filter and information from a star tracker, this method becomes a powerful tool for state estimation with minimal prior information. One challenge with actually implementing these limb-based methods is the uncertainty in the quality of measurements that will be obtained from the images. The quality of the measurements can be affected by factors such as lighting conditions, the amount of limb in the image, and distance from the body. To address this challenge, understanding the cadence of quality measurements required to obtain an accurate state estimation from an uncertain prior, as well as maintain a desired level of accuracy, is crucial. This paper explores the relationship between measurement cadence and state estimation accuracy in the context of limb-based OpNav, providing insights into how to optimize measurement strategies for deep space missions.**

I. Introduction

Deep-space navigation requires highly accurate state estimation. Unable to rely on the precision of GPS, Optical navigation techniques (OpNav) are used instead, encompassing methods such as angles-only localization, terrain-relative navigation, and horizon-based navigation [1], [2]. In particular, horizon-based methods are ideal when the spacecraft is in orbit or is entering an orbit around a celestial body and requires precise and accurate position estimation [3].

Autonomy is just as essential for deep space missions, where Earth-based orbit determination introduces costly delays that reduce valuable science collection time [3]. Because measurements are expensive in terms of data volume, mission time, and power consumption, optimizing measurement strategy is vital. Determining optimal orbital locations for high-quality measurements and the minimum measurement frequency would significantly enhance autonomous OpNav capabilities.

II. Background

A. Horizon-Based Optical Navigation (OpNav)

Horizon-based methods take an image of a celestial body's horizon (or limb), and with camera geometry and prior attitude knowledge, these methods can calculate the position of the spacecraft [4]. Photos of the limb can contain the full horizon or only a small portion, blocked by the edges of the camera frame or in shadow, giving less "edge" to work with. This navigation method has been used on missions including Artemis 1 and is planned to be used on future Artemis missions as well [5], [6].

Horizon-based optical navigation can be split into three categories. First, image processing obtains the limb/edge points of the central body. Second, the position of the satellite with respect to the central body is calculated. Finally, an EKF refines the position and obtains smoothed velocity.

B. Image Processing

The celestial body is projected onto the camera's image plane within the camera, where its apparent limb corresponds to the spacecraft's view of the horizon (visualized in Figure 1). Subscript C denotes the camera's frame and subscript P denotes the planet/body's frame.

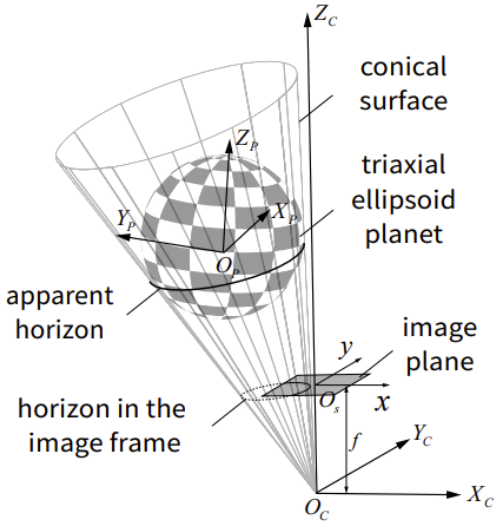


Fig. 1: Camera coordinate system and central body projection [7]

In this ideal scenario, the camera is able to determine the horizon to perfect accuracy. Realistically, error sources include

overexposure (blurry edges), memory limitations (reducing the number of pixels or precision), and image processing biases [8]. Star trackers can be used to scan images in many vertical slices to determine the approximate edge locations, only zooming in on certain windows of data for limb distinction, but edge detection may not even be accomplished if the image is overexposed [8].

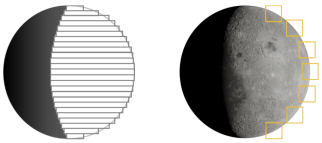


Fig. 2: Segment and Window Image Processing

C. Position Determination and the Christian-Robinson Algorithm

It is widely acknowledged that the Christian-Robinson algorithm [9], [10] is the leading algorithm for converting measurements to position information [4]. Rather than fitting an entire ellipse to the body's limb points, the formula uses a deterministic TLS (total least-squares) solution based on the output of edge detection. It transforms the image pixel coordinates to vectors towards the limb points, and vector $\boldsymbol{n}$ can be recovered as the TLS of all limb vectors, with the ellipse-transform scaling reversed to find the true r distance in the Camera Frame (shown in the Methods Section in Equation 15).

Orion's OpNav system uses the CR Algorithm for its navigation due to its non-iterative solution (with no initial guess requirement), no photometric model requirement, and fast solution. In contrast, alternative methods include using non-linear least squares (which converges in 5–10 iterations) or modeling limbs as sigmoids for robustness to blurry and over-saturated images [11]. This method is utilized by Artemis as a secondary algorithm, producing "nearly identical" results despite higher computation cost [5].

As an exact-solution method that directly considers all measurements, it is susceptible to measurement/image bias, but this can be mitigated through filtering, maintaining its appeal in low-compute spacecraft environments.

D. Previous Work and Implementation within Extended Kalman Filters

EKFs pair well with horizon-based sensing. In a previous work, Shimane and Weiss achieve kilometer-level position and cm/s-level velocity accuracy, requiring measurements only as infrequency as 12 and 48 hours [12]. However, this only tracks position and velocity. In contrast, Balossi et al. additionally track residual accelerations and image biases, sampling measurement biases from Gauss-Markov processes across 500 Monte Carlo runs [8]. They included dispersions such as star-tracker misalignment, image overexposure, and focal calibration errors, and this explicit bias inclusion improves reliability and mitigates the weakness of the widely-used Christian-Robinson algorithm.

III. Methodology

We implement a Christian-Robinson algorithm into an Extended Kalman Filter (EKF) for state estimation to test measurement strategies for a spacecraft in orbit. Simulated noise added to the pixel inputs of the CR algorithm reflects predicted image quality in the orbit based on sun position/illumination.

Our goal is to determine effective measurement strategies for accurate state estimation, starting with broad approaches that can be distilled into an optimized strategy.

A. Coordinate Frames

Two main frames were used, with a third for implementing camera specifications into our work:

1) **Camera Frame:** Z-axis along camera boresight (pointing) axis, X-axis pointing right in the image plane when viewing in boresight direction, and Y-axis down (right-handed). The origin is represented by (X_C, Y_C, Z_C) at origin O_C in Figure 1, and this frame is used by the Christian-Robinson algorithm.
2) **Planet (Inertial) Frame:** Aligned with the inertial frame, represented by (X, Y, Z) at origin O in Figure 1. Used by the propagator and EKF for state truths and estimates.
3) **Image Plane (2D):** X/Y-axes aligned with Camera Frame, Z-axis of length 1 from plane to focal point, and represented by (x, y) in Figure 1. Camera image pixels are in this frame, and are transformed to/from the Camera Frame with the Camera Calibration (K) matrix (based on camera specifications).

B. Camera Specifications

Our simplified camera model assumes no skew, using the following specs (assumed to be known to high precision).

- Focal length f: The distance from the lens optical center to the image sensor plane (e.g. $2.5\,\text{mm}$).
- Pixel spacing μ_x, μ_y: Physical X/Y spacing between pixels (e.g. $5.5\,mm/px$)
- Principal point (u_x, u_y): Z-axis intersection with the image plane (e.g. $(u_x, u_y) = (1024, 1024)\,\text{px}$ for 2048×2048 resolution).

The scaled focal lengths $d_x = f/\mu_x$ and $d_y = f/\mu_y$ are used to make the inverse Camera Calibration matrix K^{-1}, mapping a set of pixel coordinates to camera frame points [4].

$$K^{-1} = \begin{bmatrix} d_x & 0 & u_x \\ 0 & d_y & u_y \\ 0 & 0 & 1 \end{bmatrix} \tag{1}$$

Values d_x and d_y represent "unit" distances of a pixel along their respective axes. Matrices $\mathbf{K}$ and $\mathbf{K}^{-1}$ enable transformation between the image plane (pixels) and the 3D Camera Plane (meters) for both image generation (position-to-pixel) and CR estimation (pixel-to-position).

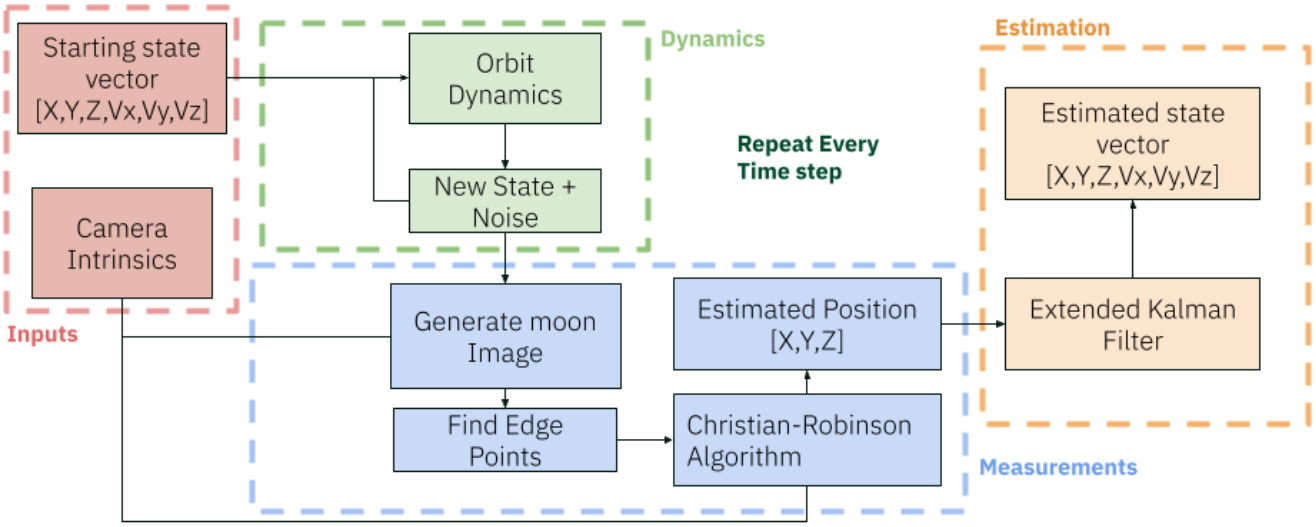


Fig. 3: Diagram of the Horizon-Based OpNav simulation pipeline showing the main areas of Inputs, Dynamics, Measurements, and Estimation

C. Orbital Dynamics

We use a two-body gravitational model in the Planet Frame with no input. Position $\mathbf{r} = [x, y, z]^T$ and velocity $\mathbf{v} = [v_x, v_y, v_z]^T$ evolve according to Newton's Universal Law of Gravitation, with μ as the gravitational parameter of the planet/central body.

In discrete time, the equations are as follows, where dt is the time delta between each step and $\mathbf{w}_k$ and $\mathbf{v}_k$ are the process and measurement noises drawn from a Gaussian distribution with covariance matrices Q_k and R_k as shown in Equations 6 and 7. The measurement of the position in the Camera Frame $\{\mathbf{r}_k\}_C$ is also derived from the Planet Frame position via the Planet-to-Camera Rotation matrix T_C^P.

$$\mathbf{r}_{k+1} = \mathbf{r}_k + \mathbf{v}_k\, dt + \frac{1}{2}\mathbf{a}_k\, dt^2 + \mathbf{w}_k \tag{2}$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{a}_k\, dt \tag{3}$$

$$\{\mathbf{r}_k\}_C = T_C^P\, \{\mathbf{r}_k\}_P \tag{4}$$

$$\mathbf{a_k} = -\frac{\mu_{cb}}{\|\mathbf{r_k}\|^3}\mathbf{r_k} \tag{5}$$

$$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0},\, Q_k) \tag{6}$$

$$\mathbf{v}_k \sim \mathcal{N}(\mathbf{0},\, R_k) \tag{7}$$

D. Image Generation

Generating synthetic body images is necessary to test the nuances of the CR algorithm. In particular, the shading of the body due to the sun was one source of error that we hoped to test by generating our own images, as this represents one of the biggest challenges to robust horizon-based OpNav. The images are generated by taking in the camera position and attitude relative to the body and the camera specifications, and projecting the geometry of the body into the camera frame. The projection is then shaded according to a fixed light source in the world frame. We then add noise to the image to make it more realistic. A generated limb image and the addition of noise are shown in Fig. 4.

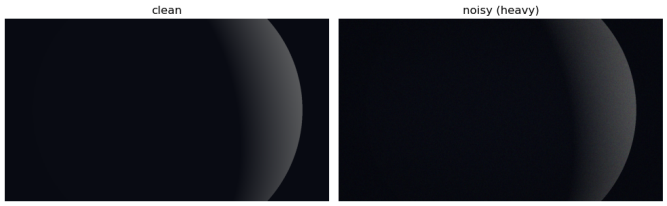


Fig. 4: Body image generation and addition of noise (right)

E. Edge Detection

In order to process the data the images provide mathematically, the edge points of the body need to be extracted. There are many valid ways of finding these edge points, but we chose to use the straightforward method of finding the highest-contrast edges and then trimming any edges that do not fall under a certain threshold.

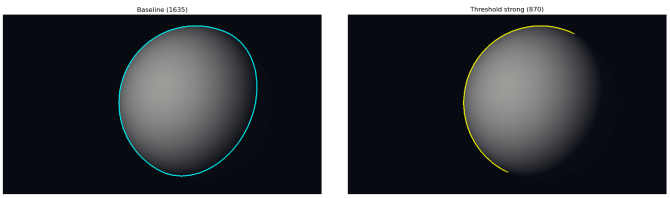


Fig. 5: Foreground masking (left) and weak edge removal (right)

- **Foreground Masking:** Extracting the object in front of a dark background by finding the points that border the

dark region.

- **Weak Edge Removal:** Removes edge points that do not have a strong edge in one direction. Fig. 5 shows the removal of weak edges.

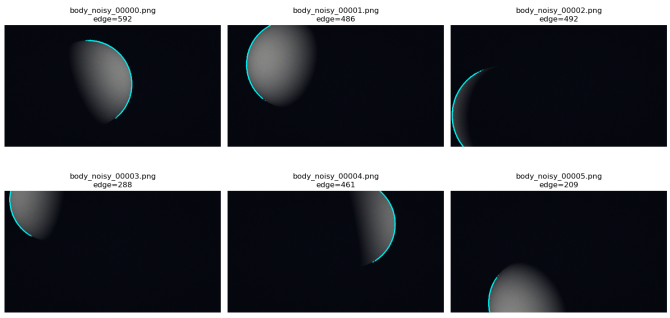


Fig. 6: Demonstration of edge detection

We added a small amount of noise to camera attitude to simulate some pointing error but otherwise we assumed the camera would point at near the center of the body.

F. Christian-Robinson Algorithm

The CR algorithm converts edge points $\boldsymbol{u}_i$ in pixel coordinates to a position estimate $\boldsymbol{r}_C$ in the Camera Frame. It takes in the Camera Calibration matrix $\boldsymbol{K}$ and the ellipsoid transformation matrix $\boldsymbol{D} = \text{diag}(a, b, c)$, with a, b, and c being the principal axes of the ellipsoidal planet in its frame. For our spherical body assumption, $\boldsymbol{D} = \boldsymbol{I}_3$.

$$\boldsymbol{D} = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \tag{8}$$

The first step is to convert $\boldsymbol{u}$ into unit vectors in the Planet Frame. This is done by using the inverse K matrix to bring the points into Camera Frame coordinates, rotating those points into the Planet Frame with $\boldsymbol{T}_P^C$ (known from the star tracker), and scaling by the ellipsoid transformation $\boldsymbol{D}$ to transform each point to the edge of a sphere of "unit radius" in the final planet frame (i.e., a tightly bounding cone).

$$\boldsymbol{R} = \boldsymbol{D}\boldsymbol{T}_P^C\boldsymbol{K}^{-1} \tag{9}$$

Then, each transformed point is scaled to get a unit "ray" vector $\boldsymbol{s}_i$, and stacked to get the matrix of all measurements $\boldsymbol{H}$.

$$\bar{\boldsymbol{x}}_i' = \boldsymbol{R}\bar{\boldsymbol{u}}_i \tag{10}$$

$$\boldsymbol{s}_i' = \frac{\bar{\boldsymbol{x}}_i'}{||\bar{\boldsymbol{x}}_i'||} \tag{11}$$

$$\mathbf{H} = \begin{bmatrix} \mathbf{s}_1'^T \\ \mathbf{s}_2'^T \\ \vdots \\ \mathbf{s}_n'^T \end{bmatrix}_{n\times 3} \tag{12}$$

This formulates a Total Least Squares (TLS) problem (Equation 13), solved using the TLS Normal Equations (Equation 14). This recovers the estimated vector towards the planet center $\boldsymbol{n}$, as pictured in Figure 7.

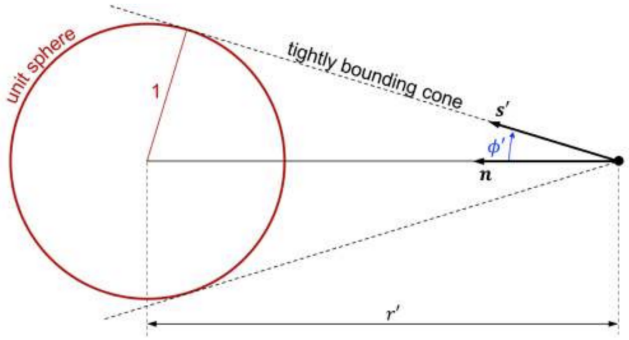


Fig. 7: Geometry of Vectors for CR Algorithm

$$\mathbf{H}\mathbf{n} = \mathbf{1}_{n\times 1} \tag{13}$$

$$\mathbf{n} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{1}_{n\times 1} \tag{14}$$

Finally, the transformations are reversed for this new vector to recover the position, and the final vector is calculated for the distance to the planet's central origin in the Camera Frame.

$$\mathbf{r}_C = (\mathbf{n}^T\mathbf{n} - 1)^{-1/2}\mathbf{T}_C^P\mathbf{D}^{-1}\mathbf{n} \tag{15}$$

G. EKF

To use the Christian-Robinson position output (in the Camera Frame) to estimate both the absolute position and velocity (Planet Frame), we implemented an Extended Kalman Filter (EKF) due to the nonlinearity of the system and the filter's historically widespread use in the space industry. While the Unscented Kalman Filter (UKF) has recently gained popularity due to faster and more efficient processors, the EKF still retains its popularity for its smaller computational requirements compared to the UKF.

The a priori state mean estimate $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$ are as follows for a given timestep t and previous timestep $t-1$, where $\mathbf{F}_{t-1}$ is the Jacobian of the state evolution function $f(\boldsymbol{\mu}_{t-1|t-1}, \mathbf{u}_{t-1})$ and $\boldsymbol{Q}$ is the process noise (Equation 6).

A Priori Estimate

$$\boldsymbol{\mu}_{t|t-1} = f(\boldsymbol{\mu}_{t-1|t-1}, \mathbf{u}_{t-1}) \tag{16}$$

$$\boldsymbol{\Sigma}_{t|t-1} = \mathbf{F}_{t-1}\boldsymbol{\Sigma}_{t-1|t-1}\mathbf{F}_{t-1}^T + \mathbf{Q}_{t-1} \tag{17}$$

With a CR measurement, the correction step follows, where $\boldsymbol{R}$ is the measurement noise (Equation 7), $\mathbf{H}_t^T$ is the Jacobian of the measurement function $h(\boldsymbol{\mu}_{t|t-1})$, and $\boldsymbol{K}_t$ is the Kalman Gain defined below.

A Posteriori (with measurement)

$$\boldsymbol{\mu}_{t|t} = \boldsymbol{\mu}_{t|t-1} + \mathbf{K}_t(\mathbf{y}_t - h(\boldsymbol{\mu}_{t|t-1})) \tag{18}$$

$$\boldsymbol{\Sigma}_{t|t} = \boldsymbol{\Sigma}_{t|t-1} - \mathbf{K}_t\mathbf{H}_t\boldsymbol{\Sigma}_{t|t-1} \tag{19}$$

$$\mathbf{K}_t = \mathbf{\Sigma}_{t|t-1}\mathbf{H}_t^T(\mathbf{H}_t\mathbf{\Sigma}_{t|t-1}\mathbf{H}_t^T + \mathbf{R}_t)^{-1} \tag{20}$$

The Jacobian of the state evolution function couples position r and velocity v through gravitational acceleration:

$$\boldsymbol{F} = \begin{bmatrix} & I_3 & I_3\,dt \\ -dt\,\mu\left(\frac{1}{\|\mathbf{r}\|^3}I_3 - \frac{3}{\|\mathbf{r}\|^5}\mathbf{r}\mathbf{r}^T\right) & & I_3 \end{bmatrix} \tag{21}$$

Since the input to the system is the position in the Camera Frame, the measurement function transforms from Planet Frame to Camera Frame, giving the Jacobian of the measurement function $h(\mu)$ as:

$$\boldsymbol{G_d} = \begin{bmatrix} T_C^P & I_3 \end{bmatrix} \tag{22}$$

The filter computes the predict step continuously, but only performs the update step when measurements are available, enabling smooth state propagation during measurement dropout/spacing to test our sparse measurement strategies.

IV. Experiments

For the experiments, we tested six different measurement strategies to investigate what is important for using horizon-based OpNav with an EKF:

- **Uniform with 10 measurements:** A total of ten measurements uniformly distributed over the trajectory.
- **Uniform with 25 measurements:** A total of twenty-five measurements uniformly distributed over the trajectory.
- **Uniform with 50 measurements:** A total of fifty measurements uniformly distributed over the trajectory.
- **Front-Loading Measurements:** A total of fifty measurements, with more measurements at the beginning.
- **Two-Batch Measurement:** A total of fifty measurements uniformly distributed over the trajectory in batches of two back-to-back measurements.
- **Distance-Based Measurements:** A total of fifty measurements distributed according to the predicted distance, with more measurements taken when closer to the body.

Visualizations of these strategies can be found in Fig. 9.

For all measurement strategies, we used a state-dependent R matrix to account for how the image noise with distance (as the same angular error will produce a much larger error when further away). The z term was also weighted higher due to a higher level of noise observed in that direction, and d_t is the distance between the spacecraft and the body.

$$R_t = \begin{bmatrix} d_t \cdot 10/3500 & 0 & 0 \\ 0 & d_t \cdot 10/3500 & 0 \\ 0 & 0 & d_t \cdot 50/3500 \end{bmatrix} \tag{23}$$

The initial R also had a coefficient multiplier of 10, which we found to be very important for the EKF to work properly. With a higher or lower value, we observed either a large lag in the estimation or divergence.

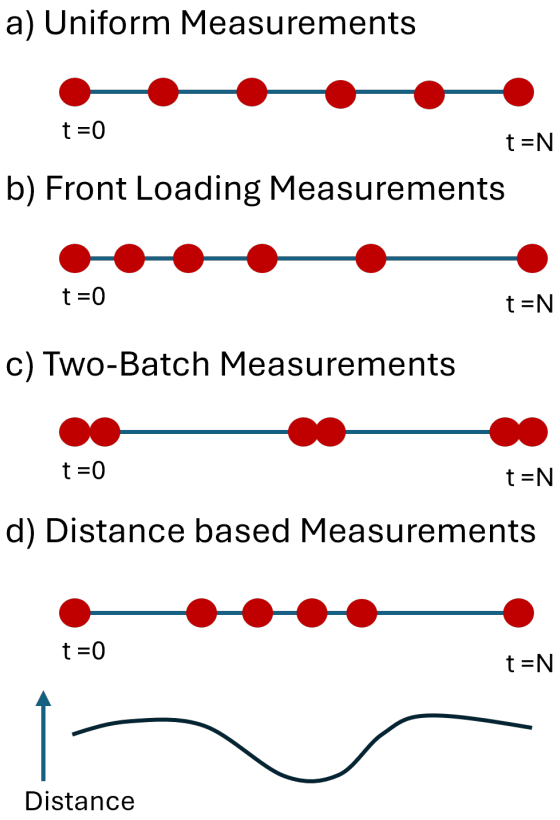


Fig. 9: Visualization of different measurement strategies, with red dots showing when a measurement is taken

We selected our orbit to start far away from the body before completing several highly elliptical orbits. This was chosen to simulate a spacecraft's approach to a body with very little knowledge of its relative or true position. Therefore, the initial EKF estimate was given a large offset. The highly elliptical orbit also provides the opportunity to test the CR algorithm with different views of the planet. The chosen initial state is:

$$\text{state}_0 = [10000, 1000, 1000, 0, 0.5, 0] \tag{24}$$

This trajectory is shown in Fig. 10 below.

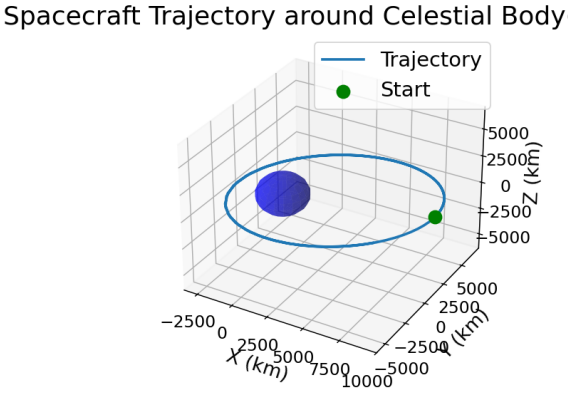
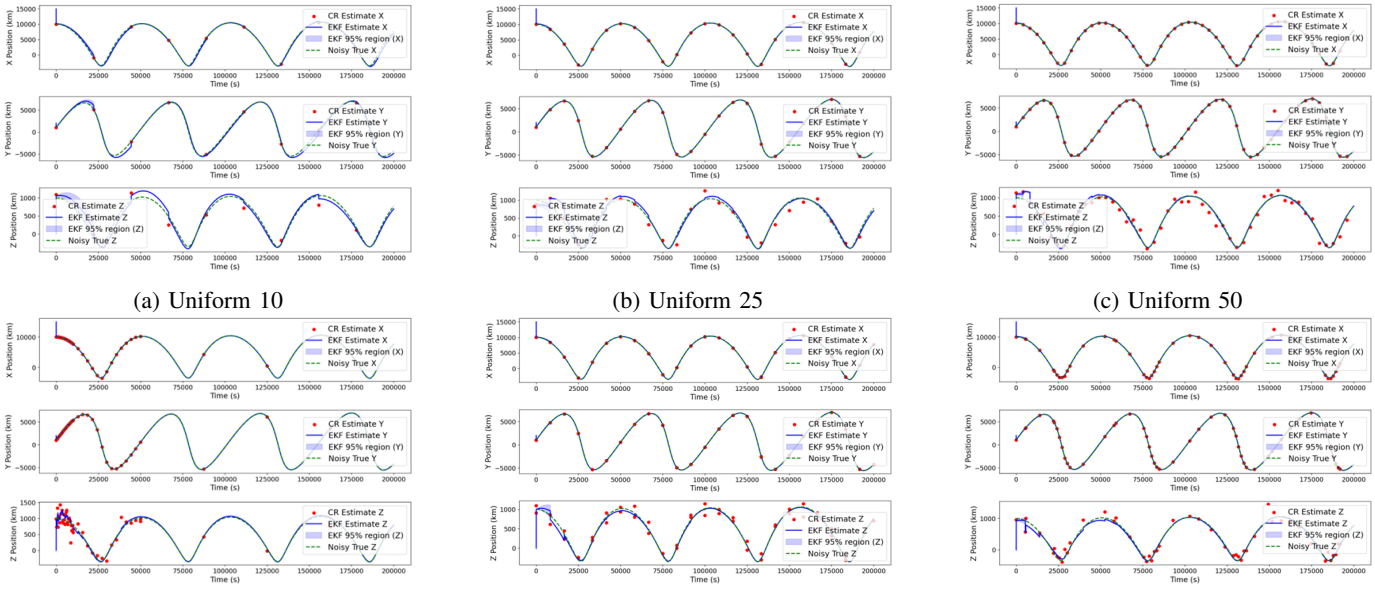


Fig. 10: Visualization of test orbit (body not to scale)



(a) Uniform 10 (b) Uniform 25 (c) Uniform 50

(d) Front loading (50 tot. meas.) (e) Batch (Two back-to-back images, 50 tot. meas.) (f) Distance-based (50 tot. meas.)

Fig. 8: X, Y, Z Position Comparison of different sampling strategies.

V. Results

In comparing the different measurement strategies, we found that comparing the Root-Mean-Squared-Error (RMSE) values at the end of the trajectory was the best way to summarize the performance of a given strategy. The X, Y, Z positions in Fig. 8 show the true noisy trajectory, the Christian-Robinson estimate (red dots), and the EKF estimate. These plots provide a good visualization of the differences in measurement strategies and where the EKF struggles to track. Most graphs show that the z direction has the greatest variance, which is also reflected in the RMSE values in Table I.

The RMSE values clearly demonstrate the advantage of increasing the measurement cadence when examining the RMSE_{3D} values for the Uniform scenarios. Specifically, using only 10 measurements yields a final three-dimensional position RMSE of 543.3 km, whereas employing 25 and 50 measurements reduces the RMSE to 133.3 km and 72.2 km, respectively. Future tests exploring the trade-off between increasing measurement cadence and RMSE could help determine what the optimal number of measurements could be.

Another key insight provided by the RMSE analysis is that the front-loading and batch-measurement strategies yield lower RMSE_{3D} values than the baseline uniform strategy for the same total number of measurements. These two strategies are motivated by more operationally realistic mission scenarios, in which maintaining a high-frequency uniform sampling schedule may be impractical. In particular, the front-loading approach is noteworthy, as it demonstrates that once a sufficiently accurate state estimate has been established, subsequent measurements can be collected at a reduced frequency without significant degradation in estimation performance. Future work could focus on combining the batch and front-loading strategies to further exploit these advantages.

TABLE I: Position RMSE Comparison Across Methods

Method	RMSE_x(km)	RMSE_y(km)	RMSE_z(km)	RMSE_{3D}(km)
Uniform (10)	378.3	380.0	99.2	545.3
Uniform (25)	72.8	46.1	101.7	133.3
Uniform (50)	44.4	16.6	55.3	72.3
Front loading	44.0	22.7	39.8	63.6
Batch	49.0	22.5	39.9	67.0
Distance-based	72.2	50.4	46.7	100.0

VI. Conclusion

This work shows that measurement strategy can make a significant difference when using horizon-based OpNav. Specifically, using a front-loaded measurement strategy and batching measurements together can improve RMSE compared to uniform measurements. As expected, a higher cadence of measurements also improves estimation.

Several limitations of the investigations presented in this paper suggest directions for future research.

1) **More realistic orbital and environmental modeling:** Applying the developed pipeline within a high-fidelity three-body dynamical framework—particularly with bodies with irregular orbits, such as about the Moon—would enable a more rigorous assessment of horizon-based OpNav as a deep space navigation technique.

2) **Incorporation of illumination conditions into measurement strategies:** Although not examined in the current study, surface illumination can significantly affect measurement quality. Integrating lighting models of the target body into a measurement strategy, and scheduling measurements at

epochs with optimal illumination geometry, may yield further improvements in state estimation accuracy.

3) **Realistic Noise and Tolerance:** Adding specific angular error and noise models will produce more realistic synthetic images and attitude estimate (which we had previously assumed to be perfectly known), especially when considering the error characteristics of varying attitude-estimation methods. For example, a typical attitude determination system might consist of a star tracker paired with gyroscopes and sun sensors, each having distinct noise characteristics and biases. By explicitly including these sensor-specific noise models, we could more improve our robustness and accuracy of the state estimation pipeline.

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